

Bridging the Skill Gap: A Bioinformatics Workshop Proposal

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Audience Analysis:

The intended audience for this proposal is the Community Engagement Committee of the Department of Cell Biology and Molecular Genetics (CBMG) at the University of Maryland College Park. This committee oversees departmental decisions like recommending and implementing activities, resources, and policies to enhance inter-departmental community engagement and growth (CBMG, n.d.). Since the CBMG committee is composed of faculty dedicated to the educational enrichment of their department, members are likely already aware of the growing gap between theoretical education and applied computational skills in the life sciences. As a bioinformatics major at a university with no formal bioinformatics program, I have experienced this gap personally and am well-equipped to properly showcase this issue and solution to the committee.

This proposal is designed to align with the committee's priorities by offering a low-cost, low-barrier, and scalable solution that does not require major curriculum restructuring. Many universities, including the University of Maryland, limit enrollment of popular programs to ensure the quality of education per departmental student doesn't decrease. Bioinformatics sits at the intersection of information science, biology, and computer science, all of which are often limited enrollment programs (LEPs) (Attwood et al., 2019). Since the workshop is offered as an optional extra-curricular activity, it bypasses many of the administrative red tape that would be involved in making LEP program curriculum adjustments. However, potential concerns may still include faculty workload and/or resource allocation. To address these concerns, the proposed workshop emphasizes use of open source technologies and freely available tools, as well as flexible staffing options and support systems. This proposal seeks to communicate clarity and feasibility to the committee via a structured, four-phased plan, while maintaining a focus on how this program can improve student outcomes that is backed by statistical evidence without compromising existing departmental resources.

Project Memo:

Undergraduate biology education programs traditionally emphasize theoretical knowledge and wet-lab based techniques, but as technology advances, the research landscape and the tools used to navigate it are changing (Boileau, 2021). Advances across high-throughput sequencing, large scale data collection, and computation modeling have drastically changed the approaches to biological research, re-categorizing computational technique education from a niche skill to a critical one (Wilson Sayres et al., 2018).

However, educational institutions have struggled to adapt at the same pace. An NCBI study in 2018 that included over 1200 U.S. institutions reported that 95% of surveyed faculty agreed with the statement: “bioinformatic education should be integrated into undergraduate life sciences education,” but only 32% reported teaching courses that cover any bioinformatics-related content (Wilson Sayres et al., 2018), which highlights both a professional acknowledgement of this systemic education gap as well as the ongoing coverage disparity in spite of said acknowledgement. This gap is in part due to the complex nature of the field; incorporating bioinformatics training requires integrating multiple disciplines, including biology, data science, computer science, and statistics, which are often administratively separated (Wilson Sayres et al., 2018). While many programs make an effort to expose students to basic programming and analysis ideas, the training fails to connect to real-world applications and processes. As a result, many graduating students possess strong conceptual knowledge but limited experience in hands-on training with computational tools and professional research practices. This disconnect is also reflected in student outcomes, with a recent study by the Society of Experimental Biology (SEB) reporting that approximately 76% of trainees are self-taught in bioinformatics skills, while only about 20% acquire these skills through formal coursework (Thompson et al., 2018). This leaves many fresh graduates unprepared for entry-level roles in biotechnology, research or other data-driven fields.

This proposal outlines the development of an optional, extra-curricular workshop/seminar series hosted by the BSCI department at University of Maryland. This program is designed to bridge the skill gap between introductory exposure and professional applications by providing hands-on training in commonly used tools, workflows, and reproducibility practices. By focusing on applied skill development in a low-burden framework, the workshop aims to improve undergraduate computational training and better prepare students for the current demands of life science professional work. Despite a reported 79% growth in the life sciences workforce over the last 20 years, industry reports from 2023 indicate that 9 out of 10 biotechnology executives either currently face or expect significant skill gaps in entry-level hires, which clearly demonstrates the broader workforce implications of this education imbalance (Ruegg, 2023).

Instructional Writing:

My instructional component will be on how to design and implement a bioinformatics workshop series. This program will comprise 4 phases: Program Objectives and Outcomes, Series Structure and Curriculum Design, Implementation Plan, and Support and Evaluation Measures.

Phase 1: Program Objectives and Target Outcomes

The first phase defines the goals and expected outcomes of the workshop series, as well as identifies the target participant population to ensure that all instructional components align with measurable skill development.

Primary Objectives:

- Bridge the gap between theoretical coursework and applied computational task practice
- Facilitate student-centered development of practical skills through interactive work sessions
- Reinforce reproducible research practices using version control and workflow pipelines
- Encourage individual skill development by providing additional topic references and resources

Expected Outcomes:

By the end of the workshop series, participants will have created a usable GitHub repository that demonstrates:

- A complete bioinformatics workflow
- Proper version control practices using Git/Github
- Documentation of methods and results for reproducibility
- Application of commonly-used bioinformatic tools
- Literacy and processing techniques for biological data file formats

Target Participants:

While there are many existing workshops that focus on exposing life science students to introductory coding and computational concepts, the largest gap in bioinformatic education is between introductory concepts and professional application (Thompson et al., 2018). This workshop series seeks to address that specific gap, targeting students that

are past introductory exposure but are not yet professionally proficient by offering practical application skills not covered in computational courses.

Therefore, the target audience must already be familiar with:

- Introductory programming concepts
- Basic command line operators and usage
- Relevant biological mechanistic background
- Commonly used statistical approaches for biological data

It is recommended that attendees either have completed or are familiar with the core concepts in the following courses: BSCI410 or equivalent, AOSC247 or INST126 or CMNS131 or equivalent, STAT464 or equivalent.

Phase 2: Series Structure and Curriculum Design

The second phase outlines the program structure, topic components, and instructional contents of the workshop series.

Program Structure:

- Duration: 7 weeks
- Frequency: 1 session/week (in person)
- Session Length: 1.5 hours
 - 1st 45 minutes: Module concept introduction and lecture
 - 2nd 45 minutes: Guided hand-on activity for application practice

Module Breakdown:

- Week 1: Introduction to -Omics Data: Sequencing Data and File Formats
- Week 2: Sequencing Alignment/ Mapping: Tools and Techniques
- Week 3: Data Analysis and Visualization Application
- Week 4: Scaling Analysis and Remote Execution: HPC + SSH
- Week 5: Workflow Systems: Implementing Pipelines (Nextflow/Snakemake)
- Week 6: Git/Github: Version Control and Reproducibility Practices
- Week 7: Pulling it All Together: End to End Pipeline in Practice

Curriculum Components:

Each module will contain:

- Lecture slides introducing key concepts and background material
- A guided activity/application practice with step-by-step instructions
- A solution key to the activity for reference and troubleshooting
- Supplemental resources for optional continued learning

Each activity portion will include operational instructions for updating for and uploading to the remote repository. Finished activities will culminate in a portfolio upon completion of the workshop series.

Phase 3: Implementation Plan

The third phase addresses logistics such as scheduling, location, instructional team staffing, technical environment requirements, and recruitment/enrollment.

Logistics:

- Hosted at an on-campus computer lab or classroom that can hold 50+ participants
- Scheduled during evening to minimize participant conflicts

Instructional Team:

Recruitment:

- Upon proposal approval, faculty recruitment will begin via department announcements, postings, and meetings

Structure:

- 1-2 lead instructors (faculty)
 - depending on availability and individual module experience/expertise
- 1-2 optional teaching assistants (advanced graduate students)
 - Act as in-session troubleshooting support and Office Hour hosts

Funding:

- Per UMD's policy, extra-curricular student engagement is a responsibility of professorships and therefore, faculty will not receive additional financial compensation for workshop involvement.

- Teaching Assistants postings will initially be posted on a volunteer basis, but financial or credit compensation options may be offered through the Undergraduate Teaching Assistant (UTA) program.

Technical Environment:

- Access to laptops with terminal capability
- Installation of required software (Git, conda, workflow packages, etc.)
- Installation and account registration of Global Protect (UMD's VPN)
- Use of free, open-source tools to minimize cost, including:
 - BLAST
 - Python
 - RStudio

Recruitment:

Advertised through:

- Community bulletin boards
- Course announcements
- Departmental mailing lists

Enrollment:

Each advertisement will include a link to the registration survey. This will include questions like:

- 1) Name
- 2) Email
- 3) Program currently enrolled in
- 4) Reason for enrolling in workshop

For the first workshop, enrollment will be capped at 50 students to ensure the instructor-to-student ratio supports accessible troubleshooting/aid for participants. Future scalability options discussed in Phase 4.

Phase 4: Support and Evaluation Measures

Phase 4 covers support structures like office hours and troubleshooting support as well as structures for evaluation, feedback, and adjustments/potential future plans.

Student Support:

- Optional office hours/drop-in troubleshooting sessions
 - Additional sessions will be offered the first week in anticipation of increased aid requests with environment setup
- Shared communication platform (Discord) will be set up and accessible for questions and discourse
- Google Drive will contain all course materials

Evaluation:

Evaluation metrics include:

- Registration survey
- Pre- and post- workshop self-assessments
- Github repository reviews
- Participant feedback surveys

These feedback methods will allow program directors to adjust the workshop curriculum, structure and logistics as needed.

Future Development:

Based on feedback, program directors could iteratively improve the workshop curriculum, structure, and logistics as needed.

Potential Expansions/Adjustments:

- Expand scale/participant count
- Expand to include online/remote sections
- Adjust curriculum in pace with technological advancement
- Additional modules

If feedback and outcomes prove significant, this workshop could eventually be integrated into a formal course.

Infographic:**Contextual Statement:**

Please see separate attachment for the infographic portion, titled “How DNA Becomes Data.” This infographic would be used in the first module, “Week 1: Introduction to -Omics Data: Sequencing Data and File Formats,” as a supplemental resource to aid in student understanding of genetic data files and how they are created. I picked this topic because this is a concept life science students often struggle with. It can be difficult to conceptualize how microscopic biological tissue samples can be digitally encoded into files, so I created an infographic that breaks the process into six simple steps. Each step contains a symbolic icon and small description to improve audience understanding and conceptual clarity on the individual and sequential steps in this complex process.

Reflective Memo:

Overall, I really enjoyed the process of this proposal assignment, and actually intend to take it further beyond course as a real proposal to the BSCI department. I initially faced some issues with section determination, and struggled to figure out what information was appropriate per segment, but I am very satisfied with the outcome after several rounds of self, peer, and instructor review. My peers were able to give me a lot of feedback, particularly on what they felt

was strong about the proposal, especially the statistical backing, flow, and workshop design. Having multiple rounds of peer feedback was really useful because it allowed me to follow up on suggestions and get input on my implementation approach for proposal changes.

While my peers did not have many concrete adjustment suggestions, my instructor was able to provide a few instances where the wording of my proposal could be improved. For example within the technical documentation in “Phase 1: Target Participants,” I softened the wording from “Recommended prerequisites:” to “it is recommended that attendees either have completed or are familiar with the core concepts in the following courses:” in order to frame concept familiarity boundaries that were more aligned with a extracurricular workshop’s expectations than a tradition course’s. This allowed me to still address prior education without engaging in the traditional course practices, like formal pre-requisites, that this workshop seeks to circumvent. The instructor also suggested re-working the wording of the final sentence of my infographic contextual statement, which really improved the clarity. What was initially a 4 line sentence was broken into two smaller and more direct statements that more accurately reflected the goals of the infographic.

I also made changes to the text chunk of the first step of the infographic, from “cells collected from target organisms, which contain DNA that can be extracted and analyzed” to “cells are collected from target organisms, which contain DNA that can be extracted and analyzed.” While this change was small, it aligned the text pattern of that chunk to the others, and improved the flow of the infographic overall. I also relocated the infographic from inside this document to a supplemental file. I did this for organizational consistency, since the additional component is also included as a separate attachment, given its format as a powerpoint is different then this proposal document. This will allow readers to more easily navigate through this proposal’s components, and creates better internal flow within the proposal document.

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